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## String :

- A string is a sequence of characters.
- A character is simply a symbol. For example, the English language has 26 characters.
- Computers do not deal with characters, they deal with numbers (binary). Even though you may see characters on your screen, internally it is stored and manipulated as a combination of 0s and 1s.
- This conversion of character to a number is called encoding, and the reverse process is decoding. ASCII and Unicode are some of the popular encodings used.
- In Python, a string is a sequence of Unicode characters. Unicode was introduced to include every character in all languages and bring uniformity in encoding.

### #Using single quotes

```
str1 = 'Hello Python'
```

```
print(str1)
```

### #Using double quotes

```
str2 = "Hello Python"
```

```
print(str2)
```

### #Using triple quotes

```
str3 = """Triple quotes are generally used for  
represent the multiline or
```

```
docstring"
```

```
print(str3)
```



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## Strings indexing and splitting

Like other languages, the indexing of the Python strings starts from 0. For example, The string "HELLO" is indexed as given in the below figure.

`str = "HELLO"`

|   |   |   |   |   |
|---|---|---|---|---|
| H | E | L | L | O |
| 0 | 1 | 2 | 3 | 4 |

`str[0] = 'H'`

`str[1] = 'E'`

`str[2] = 'L'`

`str[3] = 'L'`

`str[4] = 'O'`

## String Operators in Python

Arithmetic operators do not work on strings. But, we do have certain operators for performing special operations on strings in Python.

| Operator | Description                                                                              |
|----------|------------------------------------------------------------------------------------------|
| +        | It is used to concatenate two strings. It appends the second string to the first string. |



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|        |                                                                                                        |
|--------|--------------------------------------------------------------------------------------------------------|
| *      | It concatenates multiple copies of the same string. It is basically a repetition operator.             |
| []     | It is used for indexing. The value put between this operator returns the character at the given index. |
| [:]    | It is used for slicing. It returns the sub-strings after slicing as per the given range.               |
| in     | It returns true if the character is present in the given string.                                       |
| not in | It returns true if the character is not present in the given string.                                   |

## Slicing Strings

You can return a range of characters by using the slice syntax.

Specify the start index and the end index, separated by a colon, to return a part of the string.

### Example

Get the characters from position 2 to position 5 (not included):

```
b = "Hello, World!"  
print(b[2:5])
```

**Note:** The first character has index 0.

## Slice From the Start

By leaving out the start index, the range will start at the first character:

### Example



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Get the characters from the start to position 5 (not included):

```
b = "Hello, World!"  
print(b[:5])
```

## Slice To the End

By leaving out the *end* index, the range will go to the end:

### Example

Get the characters from position 2, and all the way to the end:

```
b = "Hello, World!"  
print(b[2:])
```

## Negative Indexing

Use negative indexes to start the slice from the end of the string:

### Example

Get the characters:

From: "o" in "World!" (position -5)

To, but not included: "d" in "World!" (position -2):

```
b = "Hello, World!"  
print(b[-5:-2])
```

## Modify Strings

Python has a set of built-in methods that you can use on strings.



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## 1.Upper Case

The upper() method returns the string in upper case:

**Example** a = "Hello, World!"

```
print(a.upper())
```

## 2.Lower Case

The lower() method returns the string in lower case:

**Example**

```
a = "Hello, World!"  
print(a.lower())
```

## 3.Remove Whitespace

Whitespace is the space before and/or after the actual text, and very often you want to remove this space.

**Example**

The strip() method removes any whitespace from the beginning or the end:

```
a = " Hello, World! " print(a.strip())  
# returns "Hello, World!"
```



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## 4. Replace String

The `replace()` method replaces a string with another string:

### Example

```
a = "Hello, World!"  
print(a.replace("H", "J"))
```

## 5. Split String

The `split()` method returns a list where the text between the specified separator becomes the list items.

The `split()` method splits the string into substrings if it finds instances of the separator:

### Example

```
a = "Hello, World!"  
print(a.split(",")) # returns ['Hello', ' World!']
```

## String Concatenation

Joining of two or more strings into a single one is called concatenation.

The `+` operator does this in Python. Simply writing two string literals together also concatenates them.

The `*` operator can be used to repeat the string for a given number of times.



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## Example

```
str1 = 'Hello'

str2 = 'World!'

# using +

print('str1 + str2 = ', str1 + str2)

# using *

print('str1 * 3 =', str1 * 3)
```

If we want to concatenate strings in different lines, we can use parentheses.

```
>>> # two string literals together
>>> 'Hello "World!'
'Hello World!'

>>> # using parentheses
>>> s = ('Hello '
...     'World')
>>> s
'Hello World'
```



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